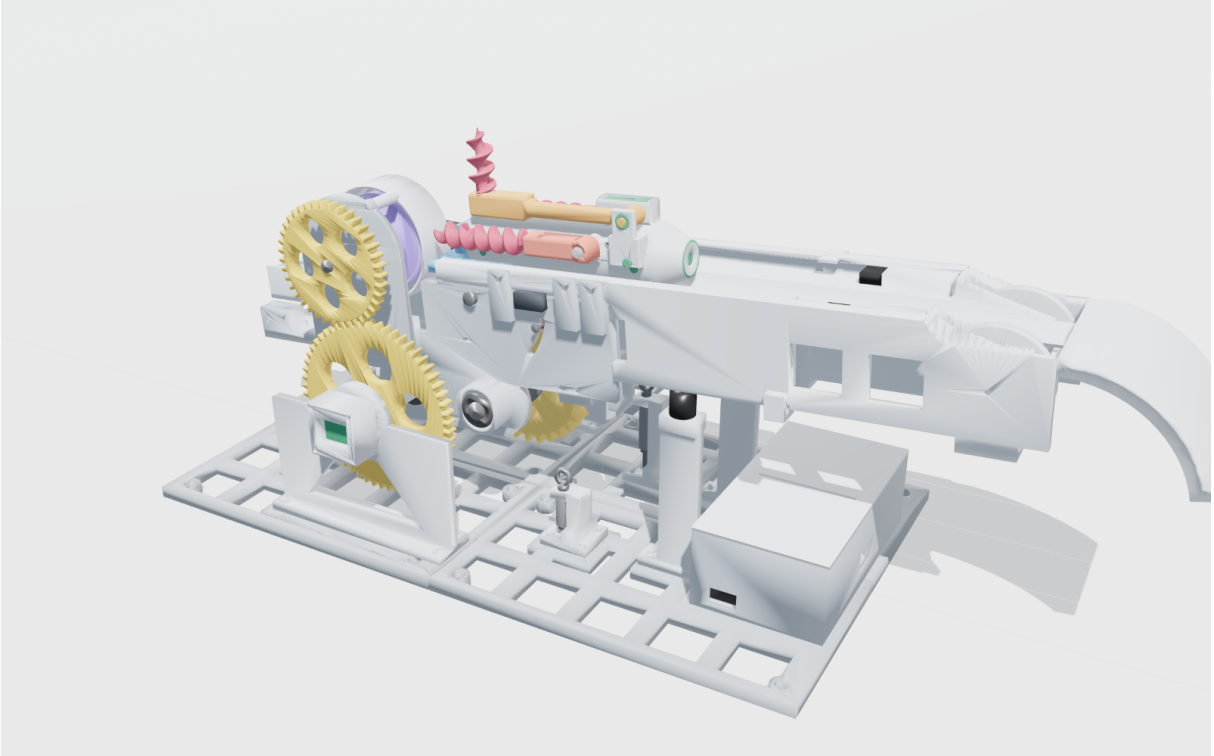
GrappStar: a rover that anchors itself and winches out

A rover stuck in sand or mud cannot push itself out. GrappStar fires a small anchor dart (KEDGE) on a tether, the dart screws itself into the ground, and a reel on the rover winches against it. Everything on the rover side runs on one motor: a 12 V car wiper motor.



1. The idea

Other approaches send a second robot to pull the first one out. GrappStar keeps it to one rover and one throw. The dart does the anchoring with its own small motors; the rover side does three jobs (launch, reel in, tow) with a single motor, using the motor's direction and two sets of one-way clutches to pick the job. Fewer motors means fewer things to fail and less mass to carry.

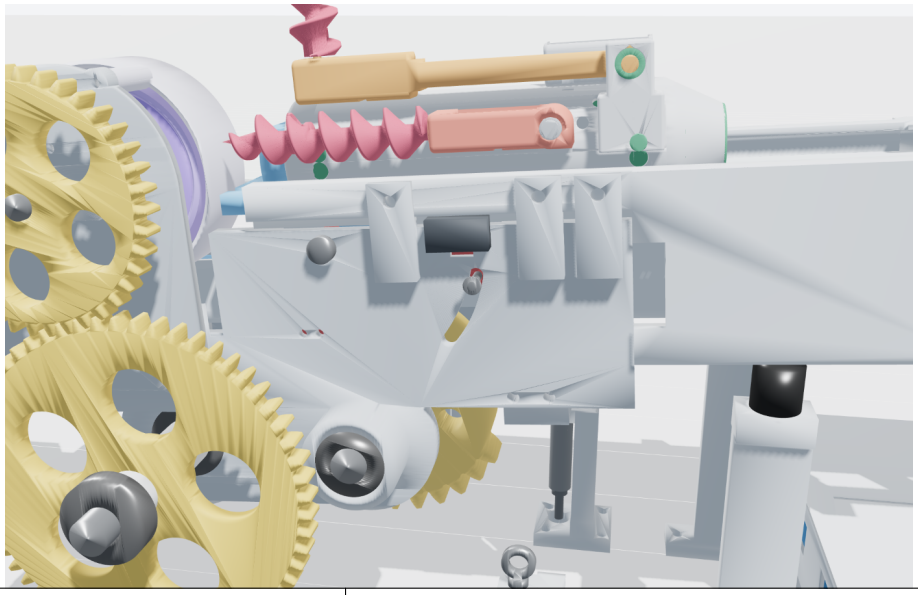
2. The dart (KEDGE v30)

A fully printed 42 mm dart, 168 mm long folded, about 279 g including 15 g of nose ballast. After landing a kick leg swings 202 deg over the nose and screws a foot auger in ahead of it, then two pods swing down 42.2 deg while two counter-rotating augers (17 mm x 72 mm, opposite hands so the dart does not twist) drill 45 mm deep. A camera in the nose sends live video to the rover; an ESP32-S3 runs the sequence and talks over WiFi.

predicted holding (Broms short pile model)	**pull before it moves**
dry sand	8 N
moist loam	48 N
firm clay	124 N

3. One motor, three jobs

Backward: a one-way clutch locks and the program gear turns once per shot. Its 18 teeth pull the carriage back 208 mm against two springs (42 to 192 N), a sear catches it, then a cam beside the gear lifts the launch rail from 4 to 35 deg. Near the top a cord tied to the base goes taut, pulls the sear down, and the dart leaves at 10.7 m/s. Forward: the launcher clutch freewheels and a backstop holds it; the reel rotor winds. The spool itself never turns: at launch the line peels off its end (like a spinning fishing reel), so there is nothing to spin up and no tangle. Stopped: the wiper's worm gear is self locking, so everything holds with the power off.



result (computed)	value
muzzle speed / range (fires at 33 deg)	10.7 m/s / 8.4 m (drag and tether included)
peak launch acceleration	46 g
peak motor load / one shot cycle	5.9 N m / 3.5 s
tow speed at 20 N	0.13 m/s
printed parts / plastic	47 parts, about 2.3 kg (gears, sear and carriage in carbon-filled nylon)

4. How it was checked before printing

- Every part is generated from one parameter file, so a change ripples through gears, cam, sear, rail and reel together.
- Collision sweep on the exact meshes: the whole mechanism stepped through the program gear's full turn in 1 deg steps (360 positions), with the dart, sear, carriage, springs and reel moving: no collisions.
- The lift cam is not drawn by hand: it is the envelope of the roller over the programmed rail angle, then checked by solving the real rail angle from the cam.
- The rack travel is found by simulating the gear teeth pushing the rack tooth by tooth, which showed the first tooth catches early and the last lets go late; the design was changed to match.
- Launch, flight and tow are computed with spring, friction, drag and tether forces; the HD simulation plays these numbers back on the real parts.
- Firmware is tested on a PC against a model of the machine (105 automatic checks) before it touches the motor.
- Six review rounds attacked the design; their findings (sear catch, trip cord, spring stress, clutches, the planet-carrier lift torque, reel and retrieve, fixings, print and assembly order) were fixed and re-checked.

5. What is still to measure

The wiper motor's real speed and torque, the bought springs' rate, and the holding force in real soil (a pull-test script logs force against movement using the rover's own reel).